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$$\mathbf{f}(x, y) = (\sin(xy), \sinh(xy))$$

$$\begin{aligned} D\mathbf{f}(x, y) &= \begin{pmatrix} \frac{\partial}{\partial x}(\sin(xy)) & \frac{\partial}{\partial y}(\sin(xy)) \\ \frac{\partial}{\partial x}(\sinh(xy)) & \frac{\partial}{\partial y}(\sinh(xy)) \end{pmatrix} \\ &= \begin{pmatrix} y \cos(xy) & x \cos(xy) \\ y \cosh(xy) & x \cosh(xy) \end{pmatrix} \end{aligned}$$

$$\det(D\mathbf{f}(x, y)) = (y \cos(xy))(x \cosh(xy)) - (x \cos(xy))(y \cosh(xy))$$

$$= xy \cos(xy) \cosh(xy) - xy \cos(xy) \cosh(xy)$$

$$= 0$$

$$\Rightarrow \det(D\mathbf{f}(x, y)) = 0 \quad \forall (x, y) \in \mathbb{R}^2$$

$$\Rightarrow \nexists^{-1}$$

References